

Road Accident Simulation Report

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Accident Type: Head-on collision

Location: Bayabas, Labo, Camarines Norte

Time of Accident: 2.94 seconds into simulation

Cause: Reckless Driving

Weather: Rainy

Road Geometry: blind_curve (Slope: 4.5°)

Vehicles Involved:

- Car 1: Mass = 1345 kg, Initial Speed = 60.0 km/h

- Car 2: Mass = 1302 kg, Initial Speed = -59.0 km/h

Analysis:

This simulation demonstrates how speed, road conditions, and lighting contribute to accidents.

Severity is classified based on impact force: minor (<50kN), moderate (50-150kN), severe (>150kN).

Recommendations:

- Maintain current lighting standards
- Improve lane markings
- Increase traffic police presence
- Enhanced driver training programs
- Weather-based speed limit adjustments

Systems Interaction Summary:

- Reckless driving combines aggressive human behavior with vehicle capabilities and environmental factors in systems theory, leading to high-risk interactions.

Validation Against Historical Records:

- Historical validation unavailable.

Intervention Scenario Comparison:

- Baseline Severity Distribution: severe: 11, none: 7, moderate: 2
- Intervention Severity Distribution: severe: 10, none: 8, moderate: 2
- Baseline Avg Impact Force: 978339.69 N
- Intervention Avg Impact Force: 725787.50 N
- Avg Risk Score Reduction: 16.4% (7.18 -> 6.00)
- Avg Impact Force Reduction: 25.8%
- Interventions Applied: Road Surface Treatment, Improved Street Lighting

Machine Learning Prediction:

Predicted Severity: Unknown (0.0% confidence)